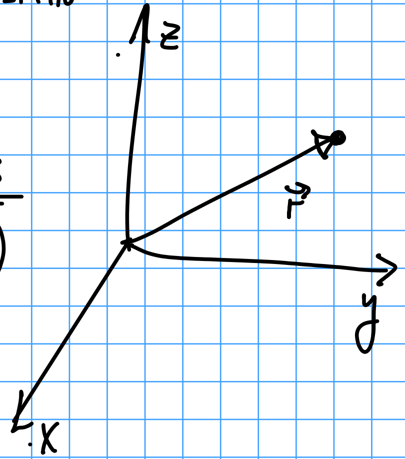
SEP 3 2024 – PHYS 6001 – Dr. H. Castillo

Mechanics of a particle

Kinematics
(DESCRIPTIVE)



$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2}{dt^2}\vec{r}$$

- Forces acting on particle

Total force (Sum of forces) $\boxed{\vec{F} = \frac{d}{dt}(m\vec{v}) = \frac{d}{dt}\vec{p} = \dot{\vec{p}}}$ $\quad \vec{p} \equiv m\vec{v}$ Linear momentum

DIFF Eqn

Newton's 2nd Law of motion

IF mass m constant $\quad \vec{F} = m\vec{a}$

THERE ARE SOME REF. FRAMES WHERE $\vec{F} = \dot{\vec{p}}$

(INERTIAL FRAME
GALILEAN FRAME)

NORMALLY, TAKE FRAME FIXED wrt EARTH $\Updownarrow$ "LAB FRAME"

- IN ASTRONOMICAL PROBLEMS, MAYBE TAKE SOME DISTANT STARS TO DEFINE FRAME
- IF TOTAL FORCE $\vec{F} = \vec{0} \Rightarrow \dot{\vec{p}} = \vec{0} \Rightarrow \vec{p}$ IS CONSERVED

- DEFINE ANGULAR MOMENTUM $\quad \vec{L} = \vec{r} \times \vec{p}$ $\quad$ (Remember $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$)

$$\dot{\vec{L}} = \frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r}\times\vec{p}) = \underbrace{\dot{\vec{r}}\times\vec{p}}_{\dot{\vec{r}}\times m\dot{\vec{r}} = 0} + \vec{r}\times\dot{\vec{p}} = \vec{r}\times\vec{F} \equiv \vec{N} \quad \text{TORQUE}$$

$(\vec{a} \parallel \vec{b} \Rightarrow \vec{a}\times\vec{b} = \vec{0})$

$$\boxed{\vec{N} = \dot{\vec{L}}}$$

- IF TOTAL TORQUE $\vec{N} = \vec{0} \Rightarrow \dot{\vec{L}} = \vec{0} \Rightarrow \vec{L}$ IS CONSERVED

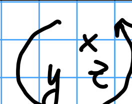
$\vec{N} = 0 \begin{cases} \vec{F} = \vec{0} \\ \vec{r} = \vec{0} \text{ (FORCES APPLIED AT THE CENTER OF ROTATION)} \\ \vec{F} \parallel \vec{r} \end{cases} \Rightarrow \vec{L}$ CONSERVED

IF $\vec{F} \parallel \vec{r} \Leftrightarrow \vec{F} = F_r\hat{r}$ $\quad \vec{N} = \vec{r}\times\vec{F} = r\hat{r}\times F_r\hat{r} = rF_r\underbrace{(\hat{r}\times\hat{r})}_{\hat{0}} = \hat{0}$

ASIDE: CROSS PRODUCTS $\quad \hat{\imath} \equiv \hat{x} \quad \hat{\jmath} \equiv \hat{y} \quad \hat{k} \equiv \hat{z}$

$\hat{x}\times\hat{y} = \hat{z} \quad \hat{y}\times\hat{z} = \hat{x} \quad \hat{z}\times\hat{x} = \hat{y}$

$\hat{y}\times\hat{x} = -\hat{z} \quad \hat{z}\times\hat{y} = -\hat{x} \quad \hat{x}\times\hat{z} = -\hat{y}$



$\vec{A}\times\vec{B} = (-\vec{B})\times\vec{A}$

$\vec{C}\times\vec{C} = (-\vec{C})\times\vec{C} = -(\vec{C}\times\vec{C}) \Rightarrow \vec{C}\times\vec{C} = \vec{0}$

Practical way to compute: use cross products of unit vectors and linearity

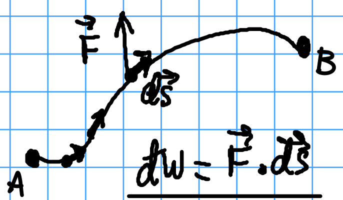
Example: $(a_x\hat{x}+a_y\hat{y})\times(b_y\hat{y}+b_z\hat{z}) = a_x b_y \underbrace{\hat{x}\times\hat{y}}_{\hat{z}} + a_y b_y \underbrace{\hat{y}\times\hat{y}}_{0} + a_x b_z \underbrace{\hat{x}\times\hat{z}}_{-\hat{y}} + a_y b_z \underbrace{\hat{y}\times\hat{z}}_{\hat{x}}$

$$= a_y b_z\hat{x} - a_x b_z\hat{y} + a_x b_y\hat{z}$$

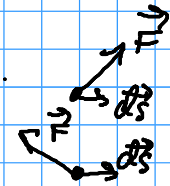
Another way: (NOT RECOMMENDED) $\vec{A}\times\vec{B} = \begin{vmatrix}\hat{x} & \hat{y} & \hat{z}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$

- WORK

$$W_{AB} = \int_A^B \vec{F}\cdot d\vec{s} = \int_A^B dW$$



$$\underline{\underline{dW = \vec{F}\cdot d\vec{s}}}$$



$dW = 0$ if $\vec{F}\perp d\vec{s}$

$$W_{AB} = \int_A^B \vec{F}\cdot d\vec{s} = \int_A^B \dot{\vec{p}}\cdot d\vec{s} = \int_A^B m\underbrace{\frac{d\vec{v}}{dt}\cdot\vec{v}\,dt}$$

$d\vec{s} = \vec{v}\,dt$

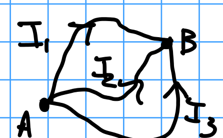
$$\frac{d\,\vec{v}\cdot\vec{v}}{dt} = \dot{\vec{v}}\cdot\vec{v} + \vec{v}\cdot\dot{\vec{v}} = 2\,\dot{\vec{v}}\cdot\vec{v}$$

$$W_{12} = m\int_A^B \frac{d}{dt}\left(\frac{v^2}{2}\right)dt = m\left.\frac{v^2}{2}\right|_A^B = \frac{m}{2}v_B^2 - \frac{m}{2}v_A^2 = T_B - T_A$$

$$\boxed{W_{AB} = T_B - T_A}$$

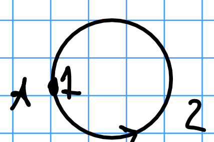
$$\boxed{\begin{array}{c}\text{KINETIC ENERGY}\\ T \equiv \frac{m}{2}v^2\end{array}}$$

$đW = \vec{F}\cdot d\vec{s}$ NOT A DIFFERENTIAL



$I_1 \neq I_2 \neq I_3 \neq I_1$ IN PRINCIPLE

EXAMPLE



$\vec{F} = \vec{F}_K$: KINETIC FRICTION

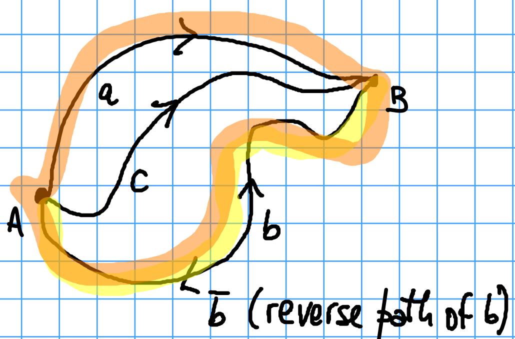
$A\to A$ 1: NO DISPLACEMENT $\Rightarrow \boxed{W_1 = 0}$

$A\to A$ 2: GO AROUND CIRCLE

$\vec{F}_K \,/\!\!/\!-\, d\vec{s} \Rightarrow dW = \vec{F}_K\cdot d\vec{s} = -|\vec{F}||d\vec{s}| < 0$

$\Rightarrow \boxed{W_2 = \int dW < 0}$

IN THIS EXAMPLE, FORCE IS NOT CONSERVATIVE

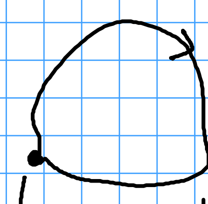


W_{12} INDEPENDENT OF PATH $\Leftrightarrow$ FORCE IS CONSERVATIVE

$\Updownarrow$

$$W_{AB}^{(a)} + W_{BA}^{(\bar{b})} = W_{AB}^{(a)} - W_{AB}^{(b)} = 0$$

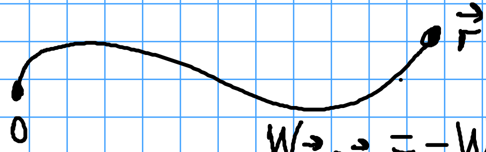
$$W = \oint_{\text{CLOSED LOOP}} \vec{F}\cdot d\vec{s} = 0$$



$$\oint \vec{F}\cdot d\vec{s} = 0$$

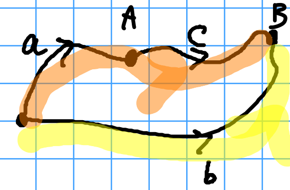
$W = 0$

FORCE IS CONSERVATIVE $\Rightarrow$ DEFINE POTENTIAL ENERGY



$$W_{\vec{r}\to\vec{0}} = -W_{\vec{0}\to\vec{r}} = V(\vec{r})$$

W_{AB}



$\left.\begin{array}{l}\text{FORCE IS}\\ \text{CONSERVATIVE}\end{array}\right) \Leftrightarrow \vec{F} = -\vec{\nabla} V \quad \underline{\underline{V = V(\vec{r})}}$

$$W_{0A}^{(a)} + W_{AB}^{(c)} = W_{0B}^{(b)} \Rightarrow W_{AB} = W_{0B} - W_{0A}$$

$$W_{AB} = V(\vec{r}_A) - V(\vec{r}_B)$$

$$W = \int_A^B -\vec{\nabla} V \cdot d\vec{r} = \int_A^B -dV = V_A - V_B$$

$$W = -\Delta V$$

$$\vec{F} \cdot d\vec{s} = -dV \qquad F_s = -\frac{\partial V}{\partial s} \qquad s \equiv \text{path length}$$

IF FORCE IS CONSERVATIVE $E = T + V$ IS CONSERVED (CONSERVATION OF MECHANICAL ENERGY)

$$T_B - T_A = W_{AB} = V_A - V_B \Rightarrow \boxed{T_B + V_B = T_A + V_A}$$

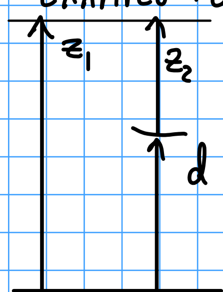
ASIDE: POTENTIAL ENERGY DEFINED UP TO A CONSTANT

$$V(\vec{r}) = -W_{\vec{r}_0\to\vec{r}} = -\int_{\vec{r}_0}^{\vec{r}} \vec{F} \cdot d\vec{s} \qquad V'(\vec{r}) = -\int_{\vec{r}_0'}^{\vec{r}} \vec{F} \cdot d\vec{s}$$

$$V(\vec{r}) - V'(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}_0'} \vec{F} \cdot d\vec{s} = V(\vec{r}_0') - V(\vec{r}_0) \quad \text{(A CONSTANT)}$$

$\left.\begin{array}{r}\text{DIFFERENT}\\ \text{STARTING POINTS}\end{array}\right\} \Leftrightarrow \left\{\begin{array}{l}\text{DIFFERENT ADDITIVE CONSTANTS}\\ \text{FOR THE POTENTIAL}\end{array}\right.$

EXAMPLE FOR GRAVITY



$$V_{grav_1} = mgz_1$$
$$V_{grav_2} = mgz_2.$$
$$V_{grav_1} = V_{grav_2} + mgd$$

<u>A CAVEAT</u>

IF $V = V(\vec{r}, t)$ and $\vec{F} = -\vec{\nabla} V \Rightarrow$ WE CAN DEFINE MECHANICAL ENERGY $E \equiv T + V$ BUT IN MOST CASES IT WON'T BE CONSERVED